Footprinting and Reconnaissance with theHarvester

Week 2 – Project Module 4 (W2-PM4) Report

Target domain: microsoft.com | Platform: Kali Linux | Tool: theHarvester

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1. Project Objective

The objective of this project is to perform passive footprinting and reconnaissance against a publicly known domain (microsoft.com) using theHarvester in Kali Linux. The project demonstrates how publicly available information such as email addresses, hostnames and subdomains can be collected from search engines and other open sources, and how results differ between a single-source search and a multi-source search.

Specific goals:

- Run theHarvester using the Baidu search engine with a result limit of 1000.
- Run theHarvester using all available sources with a result limit of 50.
- Capture screenshots and save the terminal output of both tasks.
- Compare the two approaches and document observations and learning outcomes.

2. Introduction to theHarvester

theHarvester is an open-source OSINT (Open Source Intelligence) tool included in Kali Linux. It is used during the early reconnaissance phase of a security assessment or penetration test to gather information about a target organization from public sources, without directly attacking the target's systems.

Information it can collect includes:

- Email addresses associated with the domain
- Hostnames and subdomains
- IP addresses
- Employee names and other publicly indexed data (depending on source)

It queries many data sources such as search engines (Baidu, Bing, DuckDuckGo and others) and public intelligence services (for example Shodan, VirusTotal, crt.sh). Some sources require API keys to return results.

Why footprinting matters

Footprinting is the first stage of the ethical hacking methodology. It helps security professionals understand an organization's external attack surface: what is exposed, which email formats are used, and which subdomains exist. Defenders use the same technique to discover and reduce unintended exposure before an attacker finds it.

3. Tools and Technologies Used

Item	Details
Operating system	Kali Linux (virtual machine)
Main tool	theHarvester
Interface	Linux Terminal
Utility	tee (to save output to text files)
Documentation	GitHub repository, LinkedIn post
Target	microsoft.com (public domain used for educational practice)

Command options used

Option	Meaning
-d	Target domain
-l	Maximum number of results to retrieve
-b	Data source (search engine or service)

4. Environment Setup

1. Start the Kali Linux virtual machine and log in.
2. Open a Terminal window.
3. Verify that theHarvester is installed:

```
theHarvester --help
```

4. If it is not installed, install it:

```
sudo apt update
sudo apt install theharvester
```

5. Create the project folders:

```
mkdir -p ~/Week2-PM4-theHarvester/Screenshots
mkdir -p ~/Week2-PM4-theHarvester/Results
cd ~/Week2-PM4-theHarvester
```

Screenshot placeholder: [Insert screenshot of theHarvester --help output]

5. Task 1 – Footprinting Using Baidu

5.1 Objective

Gather public information about microsoft.com using only the Baidu search engine, retrieving up to 1000 results.

5.2 Command executed

```
theHarvester -d microsoft.com -l 1000 -b baidu | tee task1_baidu.txt
```

5.3 Parameters

Parameter	Value
Target domain	microsoft.com
Source	baidu
Result limit	1000

5.4 Screenshot

[Insert task1_baidu_results.png here]

The screenshot should show the command, the target domain, the source and the results displayed.

5.5 Results and observations

- Emails found: [number / list, or "none"]
- Hosts and subdomains found: [number / list, or "none"]

- Time taken / errors or warnings: [describe]

Observation notes: [Describe what Baidu returned. A single search engine often returns a limited set of results, and may return few or no emails and hosts for a large, well-protected domain. State honestly what you saw.]

6. Task 2 – Footprinting Using All Sources

6.1 Objective

Gather public information about microsoft.com using every source supported by theHarvester, with a lower limit of 50 results per source.

6.2 Command executed

```
theHarvester -d microsoft.com -l 50 -b all | tee task2_all_sources.txt
```

6.3 Parameters

Parameter	Value
Target domain	microsoft.com
Source	all
Result limit	50

6.4 Screenshot

[Insert task2_all_sources_results.png here]

6.5 Results and observations

- Emails found: [number / list, or "none"]
- Hosts and subdomains found: [number / list]
- IP addresses found: [number / list, if any]
- Sources that failed or required API keys: [list any warnings]

Observation notes: [Describe the variety of data returned. Using all sources usually produces a broader set of subdomains and hostnames because different services index different data. Note that results can differ from the course PDF because public data changes over time, and some sources may be unavailable or need API keys in your installed version.]

7. Comparison of Both Tasks

Aspect	Task 1	Task 2
Source	Baidu	All sources
Result limit	1000	50
Number of sources queried	One	Many
Output file	task1_baidu.txt	task2_all_sources.txt
Emails found	[fill in]	[fill in]
Hosts / subdomains found	[fill in]	[fill in]
Runtime	[fill in]	[fill in]

Analysis

- **Breadth versus depth:** Task 1 requests many results from a single engine, while Task 2 requests fewer results from each of many sources. The second approach generally gives broader coverage of different data types.
- **Reliability:** A single source depends on what that engine has indexed. Multiple sources reduce blind spots but may produce errors if some require API keys.
- **Time and noise:** Querying all sources takes longer and can produce more warnings, but gives a fuller picture of the external footprint.
- **Practical takeaway:** In a real assessment, analysts start broad, then narrow down to the most productive sources and verify findings manually.

8. Security Implications

Publicly discoverable emails and subdomains can be misused for phishing, social engineering and targeting forgotten or poorly secured systems. Organizations can reduce risk by:

- Monitoring their own exposure with OSINT tools regularly
- Decommissioning unused subdomains and DNS records
- Limiting email addresses published on web pages
- Training staff to recognize phishing
- Applying strong email security controls (SPF, DKIM, DMARC)

9. Ethical and Legal Notes

This exercise used only passive, publicly available information gathering, and the target was used purely for educational practice as specified in the course module. theHarvester should only be used against domains you own or have written permission to assess. The findings in this report have not been used to access or interfere with any system.

10. Learning Outcomes

- Understood the purpose of footprinting and passive reconnaissance in ethical hacking.
- Learned to use theHarvester with the -d, -l and -b options.
- Learned how different data sources affect the quantity and type of results.
- Practiced saving evidence with screenshots and the tee command.
- Practiced organizing a project folder and documenting work in a GitHub repository.
- Understood how exposed public information can increase an organization's attack surface.

11. Conclusion

This project demonstrated how theHarvester can collect publicly available information about a domain using both a single search engine (Baidu) and all supported sources. Comparing the two tasks showed that source selection strongly affects the amount and variety of data discovered, and that results vary over time and by tool configuration. The exercise reinforced that reconnaissance is a foundational step in security testing and that organizations should regularly review what their public footprint reveals. [Add one or two sentences about your own key finding.]

12. Project Files

```
Week2-PM4-theHarvester/
├── README.md
├── Report/
│   └── theHarvester_Report.pdf
├── Screenshots/
│   ├── task1_baidu_results.png
│   └── task2_all_sources_results.png
└── Results/
    ├── task1_baidu.txt
    └── task2_all_sources.txt
```

GitHub repository: [Paste your repository link]

13. References

- theHarvester project documentation (GitHub: laramies/theHarvester)
- Kali Linux tools documentation (kali.org/tools/theharvester)
- Course material: W2-PM4 – Footprinting with theHarvester